

1.3 Computer networks, connections and protocols

Where this unit sits, what it covers, and how far

Where this unit sits

Qualification	GCSE Computer Science (9-1), specification code J277, from 2020
Component	Component 01: Computer systems — 80 marks, 1 hour 30 minutes
What OCR says the component covers	the central processing unit (CPU), computer memory and storage, data representation, wired and wireless networks, network topologies, system security and system software
This unit	topic 1.3 Computer networks, connections and protocols
Checked	3 October 2026, at https://www.ocr.org.uk/qualifications/gcse/computer-science-j277-from-2020/specification-at-a-glance/

How far this has been checked. The topic numbers and titles below are the ones this course uses on its worksheets and slides. They were not re-checked against the published specification document when this map was built: the document could not be retrieved from this build environment. Check them against <https://www.ocr.org.uk/Images/558027-specification-gcse-computer-science-j277.pdf> before relying on this map for a scheme of work, and treat any finer subdivision as absent rather than assumed - no sub-point numbers are claimed anywhere in these documents, because none were verified.

Before it: 1.1 Systems architecture, 1.2 Memory and storage

After it: 1.4 Network security, 1.5 Systems software, 1.6 Ethical, legal, cultural and environmental impacts of digital technology, 2.1 Algorithms, 2.2 Programming fundamentals, 2.3 Producing robust programs, 2.4 Boolean logic, 2.5 Programming languages and Integrated Development Environments, PY Python programming

The lesson sequence and the questions it answers

#	LESSON	THE BIG QUESTION	ACTIVITIES	MARKS
1	Types of network: zoom out nw-101	What do you gain, and what do you give up, by connecting a computer to a network?	27	35
2	Network control room nw-102	Why is the same network fast at one moment and slow at another?	14	23
3	Network showdown nw-103	What are the differences between peer-to-peer and client-server networks?	13	26
4	Mission: wire up the school nw-104	What hardware does it take to turn a room full of computers into a network?	13	16
5	Inside the internet nw-105	How does the internet work?	13	27
6	Star and mesh networks nw-106	Which shape of network would you fit here, and what are you giving up by choosing it?	22	32

#	LESSON	THE BIG QUESTION	ACTIVITIES	MARKS
7	Revision HQ nw-107	Which parts of this topic could I explain to somebody else, and which can I only recognise when I see them?	19	23
8	Assessment 1 nw-108-test	assesses the whole unit	0	0
9	Connection café nw-109	Which is better: a wired or wireless network?	13	28
10	Cipher room nw-110	What is the purpose of encryption?	13	24
11	Address sorting office nw-111	What are the differences between the three types of network address?	13	21
12	Protocol city nw-112	What are standards and protocols?	14	28
13	Parcel depot nw-113	What are the benefits of layering protocols?	12	25
14	Assessment 2 reflection nw-114-test	assesses the whole unit	0	0
14	Revision HQ 2 nw-114	Which parts of this topic could I explain to somebody else, and which can I only recognise?	20	30

Prior knowledge, vocabulary and skills

Assumed before the unit: the outcomes of 1.1, 1.2

Vocabulary introduced: Standalone computer, LAN, WAN, Bandwidth, Latency, Transmission media, Error rate, Client-server, Peer-to-peer, Switch, Router, Network interface card, Fibre optic cable, DNS (Domain Name Service), Hosting, The cloud, Topology, Star, Full mesh, Partial mesh, Ethernet, Wi-Fi, Bluetooth, Encryption, Public key, Private key, MAC address, IP address, Standard, Protocol, Layer

What a pupil does, not only reads: match, mcq, multi, order, sort

Coverage matrix

Every outcome this unit states, and whether it is taught, practised and assessed **within this unit**. Full means all three; partial means some; not covered means none of the three could be traced.

How this was worked out. The trace is derived by matching the content words of an outcome against the station text, the activities and the exam questions of the unit, not authored by hand. The words each match was made on are in build/coverage.json. It shows where to look; it does not certify that the alignment is right, and a full row has not been checked by a person either.

OUTCOME	LESSON	TAUGHT	PRACTISED	ASSESSED	COVERAGE
know what is meant by a standalone computer nw-101-o1	nw-l01	yes	yes	yes	Full
know the different types of network: LAN and WAN nw-101-o2	nw-l01	yes	yes	yes	Full
understand the advantages of networking nw-101-o3	nw-l01	yes	yes	yes	Full

OUTCOME	LESSON	TAUGHT	PRACTISED	ASSESSED	COVERAGE
understand the implications (disadvantages) of networking nw-101-o4	nw-l01	yes	yes	yes	Full
understand what factors can affect the performance of a network nw-102-o1	nw-l02	yes	yes	yes	Full
be able to explain how each factor affects network speed nw-102-o2	nw-l02	yes	yes	yes	Full
know what a client-server model is nw-103-o1	nw-l03	yes	yes	yes	Full
know what a peer-to-peer model is nw-103-o2	nw-l03	yes	yes	yes	Full
understand the different roles computers have in each model nw-103-o3	nw-l03	yes	yes	yes	Full
know the hardware needed to connect a LAN nw-104-o1	nw-l04	yes	yes	yes	Full
understand the purpose of each piece of hardware and the jobs it does nw-104-o2	nw-l04	yes	yes	yes	Full
understand what the internet actually is nw-105-o1	nw-l05	yes	yes	yes	Full
understand the terms Domain Name Service and domain name server nw-105-o2	nw-l05	yes	yes	yes	Full
understand what is meant by hosting and the cloud nw-105-o3	nw-l05	yes	yes	yes	Full
understand what is meant by the terms web server and client nw-105-o4	nw-l05	yes	yes	yes	Full
know what a star network is nw-106-o1	nw-l06	yes	yes	yes	Full
know what a mesh network is nw-106-o2	nw-l06	yes	yes	yes	Full
understand that the internet is an example of a partial mesh network nw-106-o3	nw-l06	yes	yes	yes	Full
know the advantages and disadvantages of star and mesh networks nw-106-o4	nw-l06	yes	yes	yes	Full
know which lessons I am secure on and which I need to revise nw-107-o1	nw-l07	yes	yes	yes	Full
have completed any outstanding work from lessons 1 to 6 nw-107-o2	nw-l07	yes	yes	yes	Full
be ready for my mid-topic test next lesson nw-107-o3	nw-l07	yes	yes	—	Partial

OUTCOME	LESSON	TAUGHT	PRACTISED	ASSESSED	COVERAGE
understand that Ethernet is a wired method of connection nw-109-o1	nw-l09	yes	yes	yes	Full
understand that Wi-Fi and Bluetooth are wireless methods of connection nw-109-o2	nw-l09	yes	yes	yes	Full
understand the benefits and drawbacks of wired versus wireless connections nw-109-o3	nw-l09	yes	yes	yes	Full
be able to recommend a connection type for a specified scenario nw-109-o4	nw-l09	yes	yes	yes	Full
know the basics of how cryptography can work with a simple key nw-110-o1	nw-l10	yes	yes	yes	Full
know how wireless devices authenticate each other before communicating data nw-110-o2	nw-l10	yes	yes	yes	Full
understand the difference between private and public keys nw-110-o3	nw-l10	yes	yes	yes	Full
understand why private keys are never shared nw-110-o4	nw-l10	yes	yes	yes	Full
understand the uses of MAC and IP addressing nw-111-o1	nw-l11	yes	yes	yes	Full
understand the difference between IPv4 and IPv6 nw-111-o2	nw-l11	yes	yes	yes	Full
understand the need for IPv6 nw-111-o3	nw-l11	yes	yes	yes	Full
understand the need for standards in computing nw-112-o1	nw-l12	yes	yes	yes	Full
understand the seven common protocols and what they are used for nw-112-o2	nw-l12	yes	yes	yes	Full
know why protocols are layered nw-113-o1	nw-l13	yes	yes	yes	Full
understand the advantages of layering nw-113-o2	nw-l13	yes	yes	yes	Full
know which lessons I am secure on and which to revise nw-114-o1	nw-l14	yes	yes	—	Partial
be ready for the end-of-unit assessment nw-114-o2	nw-l14	—	—	—	Not covered

36 full · 2 partial · 1 not covered, of 39 outcomes.

Gaps, assumptions and overlap

None found by the trace.

What could not be traced

The wall text of nw-102, nw-107 is painted into the 360 image and held nowhere as text, so an outcome in those lessons lesson can only be traced through its activities and its exam questions. Read the rows above for them with that in mind.

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